

Barangay VMS

Defense Q&A: Tricky Questions, Edge Cases & Scenarios

Comprehensive Q&A for Capstone Project Defense

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TRICKY & EDGE CASE QUESTIONS

Q1: A vehicle passes a Condition Check as "Good" but fails a Readiness Check. Why is this not a contradiction?

Answer: They measure completely different things and operate independently.

Condition Check asks: "Is the vehicle mechanically damaged or worn?" Good/Needs Inspection/Needs Repair/Damaged.

Readiness Check asks: "Can this vehicle deploy right now?" Fuel tank full / Water tank full / Hoses aboard / Lights work, etc.

Real example: Ambulance Doss has perfect brakes, perfect engine, no rust, no leaks → Condition = Good. But the oxygen tank was used yesterday and never refilled. Readiness = Not Ready. The vehicle is mechanically sound (Good condition) but operationally unprepared (Not Ready). This is exactly the gap the system is designed to catch.

Why this matters: A dispatch officer cannot tell by looking at a vehicle whether it's ready to deploy. The readiness check forces someone to actually verify fuel, water, hoses, equipment before the vehicle leaves the station.

Q2: Can a Maintenance Ticket close even if a sub-issue is still "Under Repair"?

Answer: Yes, but only through "Decision Close" and with explicit conditions.

There are two ways to close a ticket:

1. Clean Close (all sub-issues Done) → Button enabled, one click, vehicle returns to Available.

2. Decision Close (unfinished sub-issues) → Admin must provide:

- Deferral reason (e.g., "Part backordered 3 weeks")
- Fit-for-service verdict: "Fit" or "Not Fit"

Example: Ticket has 2 sub-issues: Front Pads (Done), Rear Pads (Under Repair, backordered).

Admin closes with reason "Part on backorder" and verdict "Not Fit". Result:

- Ticket → Closed (in archives)
- Rear Pads sub-issue → Deferred
- Vehicle → Under Maintenance (stays locked, cannot dispatch)
- Issue Report breadcrumb → Created, so defect isn't forgotten

If Admin had answered "Fit for service" instead, vehicle would return to Available, but the breadcrumb report would still exist as a reminder to complete the rear pads later.

Q3: What's the difference between a Custodian rejecting a repair and an Admin reopening it?

Answer: Same outcome (sub-issue returns to Under Repair), but different gates and contexts.

Custodian Rejection (Verify step):

- Custodian test-drives the vehicle, runs functional checklist
- If any checklist item = Fail → Approved button is disabled entirely
- Custodian must choose Rejected with notes
- Sub-issue → Under Repair, mechanic notified to redo
- Mechanic's original repair log is discarded/superseded

Admin Reopen (Confirm step):

- Custodian already approved (all checklist items passed, attestation signed)
- Admin reviews custodian's notes, mechanic's work log, receipts
- Admin disagrees with the approval (maybe feels rushed, or cost seems wrong)
- Clicks Reopen instead of Confirm
- Sub-issue → Under Repair, mechanic notified, custodian's verification is deleted

Real scenario: Custodian fails the checklist because brakes still feel mushy → Rejected immediately. VS. Custodian passes checklist and approves, but the parts receipt shows \$1000 for pads that should cost \$200 → Admin reopens to investigate the cost discrepancy.

Rejection is about the vehicle not working. Reopening is about Admin second-guessing the process or cost.

Q4: How does the system prevent two identical tickets from being created on the same vehicle?

Answer: Duplicate detection by Main Issue title.

When Admin creates a ticket, the system checks:

- Does this vehicle have any Open or Active ticket with the same Main Issue title?
- If YES → Block creation with message: "An open ticket titled '[Title]' already exists on this vehicle (Ticket #[ID]). Please use the existing ticket or close/cancel it first."
- If NO → Allow creation

Example: Roel creates Ticket #1234 for Firetruck 2 titled "Engine Overheating". Later he tries to create another ticket for Firetruck 2 with the same title. System blocks it.

Why this matters: Without this check, two mechanics could be working on the same defect unknowingly, or an admin could forget they already opened a ticket and create a duplicate, wasting time and creating confusion in the audit trail.

The system compares the exact Main Issue title (case-sensitive), not vehicle + work type. So "Engine Overheating" and "Engine Overheating - needs inspection" are different titles and both can exist. This is intentional — they're genuinely separate issues.

Q5: What happens if a Mechanic is reassigned mid-repair (they get sick, they quit)?

Answer: Admin can reassign the work order to a different mechanic without losing context.

Process:

- Sub-issue is Under Repair, assigned to Toto
- Toto becomes unavailable (sick, quit, transferred)
- Admin opens the sub-issue → Reassign Mechanic
- Selects new mechanic (Marco) + provides required reason ("Original mechanic unavailable due to illness")
- Sub-issue → still Under Repair (no status change)
- Assigned mechanic changes from Toto to Marco
- Both mechanics notified (Toto: "Work reassigned", Marco: "New work order assigned")

What happens to Toto's work? If Toto already logged some work (e.g., "Removed old pads, waiting for new parts to arrive"), that partial work log remains as context. Marco can see what was done and continue from there. When Marco completes the repair and logs final work, both Toto's partial and Marco's completion are visible in the full history.

Real scenario: Toto was assigned to replace brakes and had already removed the old pads (logged as "partial work"). He gets in a car accident and is hospitalized. Admin reassigns to Marco. Marco can see what Toto did (old pads are off, waiting for new ones), and continues from that point. Final repair log shows both contributions.

Q6: Why does Readiness Check expire but Condition Check doesn't?

Answer: They decay at different rates in the real world.

Condition Check (permanent until replaced):

- Checks for damage, wear, rust, leaks — things that change slowly over days/weeks
- A dent doesn't heal overnight; a vehicle checked "Good" today is still mechanically sound tomorrow
- New check only replaces the old one when something actually changed
- No expiration needed because the state it measures is stable

Readiness Check (expires, e.g., 7 days):

- Checks for consumables and loadout: fuel, water tank, hoses, stretcher, oxygen
- These change hourly — a full tank on Monday is half-empty after a call on Wednesday
- A check that passed 10 days ago might fail today without needing a "new" event to trigger checking
- Expiration forces periodic re-verification (every 7 days) to catch drift that happened between checks

Example: Firetruck 2 checked "Good" condition on July 1 (no damage). July 23 it's still Good, no re-check needed. BUT: readiness was checked July 15, passed. July 23 it shows "Stale" even if nothing changed, because 8 days exceeded the 7-day freshness window. Mark must re-run to confirm it's still ready or catch that the fuel tank is now half-empty.

Q7: Can an Issue Report exist without a Maintenance Ticket?

Answer: Absolutely. They are separate entities with different lifespans.

Issue Report lifecycle:

- Created by Custodian/anyone reporting a defect
- Status = Pending (waiting for Admin review)
- Admin can: (1) Reject/delete it, (2) Create a Ticket from it, (3) Ignore it and leave it Pending
- Only moves to Resolved when a linked Ticket is closed

Ticket creation is optional: Admin might review an Issue Report and decide "This doesn't need a formal ticket" (false alarm, already fixed informally, not severe enough). They can delete/close the report without opening a ticket.

Example: Nicole files an Issue Report: "Brake pedal felt soft on Tuesday." Roel reviews and calls Nicole. She confirms she may have been tired — she test-drives it again Friday and brakes feel normal. Roel deletes the report. No ticket needed.

Conversely: Ticket can exist without an Issue Report. Admin can directly create a ticket (Maintenance Tickets → Add) without any prior report. No Issue Report = no Resolved marking at the end, just a closed ticket.

Issue Reports are breadcrumbs for potential work. Tickets are formal work orders. One doesn't require the other.

Q8: What happens if a part is backordered and will arrive in 3 weeks? How does the system handle this?

Answer: Admin defers the sub-issue, and the system creates a breadcrumb to never forget about it.

Scenario: Ticket has 2 sub-issues: Front Pads (Done), Rear Pads (Open, part backordered).

Option 1: Defer now, create new ticket later

- Admin defers Rear Pads with reason: "OEM rear pads backordered from supplier, ETA July 30"
- Sub-issue → Deferred (terminal state)
- Issue Report breadcrumb → Created: "Rear Brake Pads — deferred from Ticket #1234"
- Admin closes the main ticket (it can close now with deferred work)
- Vehicle returns to Available (or Not Fit if Admin chooses)
- July 30 when parts arrive: Admin can create a new ticket from the breadcrumb report OR reopen the original

Option 2: Keep sub-issue in "Under Repair", mechanic waits

- Mechanic hasn't logged work yet (still waiting for parts)
- Sub-issue stays Open or Under Repair (depending on when mechanic was assigned)
- Ticket cannot close until this sub-issue progresses (Done or Deferred)
- Vehicle stays Under Maintenance the whole time

Why Option 1 (defer) is better: Frees the vehicle back to Available. The defect is still on record (Issue Report breadcrumb). No risk of the work being forgotten. When parts arrive, it takes 30 seconds to create a new ticket from the breadcrumb.

Q9: How does the Deferral system prevent a defect from being silently forgotten?

Answer: Deferral forces creation of a permanent Issue Report breadcrumb that acts as a reminder.

When Admin defers a sub-issue:

- Sub-issue status → Deferred (cannot be reopened or edited)
- System automatically creates or resurfaces an Issue Report
- That Issue Report has status Pending and title includes "(deferred from Ticket #[ID])"
- The deferral reason is recorded in both the sub-issue and the Issue Report

Example: Ticket #1234 on Ambulance Doss had two sub-issues: Front Pads (Done), Rear Pads (Deferred because parts backordered). Admin defers with reason: "OEM parts ETA 3 weeks."

System automatically creates Issue Report titled "Rear Brake Pads — deferred from Ticket #1234". Status = Pending.

What happens next:

- Weeks pass. Someone might ask "Are there any pending defects on Ambulance Doss?"
- They can search Issue Reports and find the Rear Pads report still marked Pending
- They cannot miss it unless they explicitly delete it
- When parts arrive, they can create a new Ticket from that Issue Report (one click)

Without the breadcrumb system, a deferred sub-issue could be lost in closed ticket archives and never reworked. The Issue Report breadcrumb ensures it stays visible in the active Issues list.

Q10: Can a non-assigned Custodian inspect a ticket?

Answer: No. Permission check blocks it.

Scenario: Ticket #1234 is assigned to Nicole for inspection. Mark (a different Custodian) opens the ticket and tries to click the Inspect button.

System response: "You are not the assigned custodian for this ticket. Only Nicole can run the inspection."

Why this check exists:

- Accountability — one person per role owns each step
- Prevents accidental double-work (two custodians both inspecting the same ticket independently)
- Audit trail clarity — if something is wrong with the inspection, we know exactly who did it

If Nicole is unavailable: Admin can cancel her assignment and reassign to Mark, but she cannot be work-sharing the inspection. One person owns it.

Q11: What is the attestation checkbox on the Verify Repair form, and why is it mandatory?

Answer: It's a signed affidavit that the Custodian personally tested the vehicle. Mandatory for approval.

The checkbox: "I have personally tested this vehicle and verified the repair myself."

Why mandatory:

- Without it, a Custodian could click Approve without actually test-driving the vehicle
- Creates legal/contractual evidence of physical verification
- Prevents rubber-stamping approvals
- Audit trail shows: "on [date] at [time], [person] personally attested to testing this vehicle"

Scenario: Nicole verifies front brake repair. Marks all checklist items Pass. Tries to click Approve without checking the attestation box. System blocks: "You must attest that you have personally tested this vehicle." She checks the box, then Approve is enabled.

What if she clicks Approve but didn't actually test? That would be fraud and would be discoverable in the audit log. If brakes fail later and someone checks, they can see Nicole's attestation and know to follow up with her.

Q12: How does the dashboard detect a "single-point-of-failure" (fragility)?

Answer: By counting ready units per vehicle type and alerting when it drops to 1 or 0.

Calculation:

- System groups vehicles by type (Ambulance, Fire Truck, Rescue Boat, etc.)
- For each type, counts: $\text{[Ready]} / \text{[Total]}$
- If Ready count = 1 → "Fragile — one unit away from zero coverage"
- If Ready count = 0 → "No coverage — cannot respond to this type of call"

Example: Barangay has 2 ambulances: Ambulance Doss (Available/Good/Ready) and Ambulance Zero (Under Maintenance/Needs Repair/Not Ready). Readiness count: 1/2. Dashboard shows Ambulances with a "Fragile" badge.

If Doss then breaks down and goes Under Maintenance, Readiness: 0/2. Badge changes to "No Coverage — Critical".

Why this matters: Roel can see at a glance: "If Ambulance Doss fails now, we have zero ambulances." This informs decisions like: prioritize repairs on Doss, or schedule preventive maintenance on Zero during off-peak hours only.

Q13: A vehicle is "Under Maintenance" but has no active ticket. Is this a bug?

Answer: No, it's expected behavior from Condition Checks and Maintenance Records.

Causes:

- **Condition Check:** Someone logs a Condition Check with result "Needs Repair" or "Damaged" → vehicle status immediately flips to Under Maintenance, no ticket required
- **Standalone Maintenance Record:** Someone creates a Maintenance Record directly (shortcut for quick repairs) → vehicle → Under Maintenance while the record is under verification/confirmation
- **Closed Ticket with "Not Fit" Decision:** Ticket is closed but Admin answered "Not Fit for service" → vehicle stays Under Maintenance

Example: Mark does a Condition Check and finds rust developing → logs "Needs Repair". Ambulance Doss now shows Under Maintenance. No ticket opened yet (Admin hasn't reviewed it). This is fine — Doss is flagged as needing attention, but the formal ticket process hasn't started.

How to fix it: Admin can either (1) create a ticket to formally address it, or (2) if it was a false alarm, Admin logs a new Condition Check as "Good" to release it.

Q14: What's the difference between "No Issues" found during inspection vs. "Needs Maintenance"?

Answer: Completely different outcomes for the vehicle and the ticket.

"No Issues" inspection:

- Custodian physically inspects, finds nothing wrong
- Submits result "No Issues"
- No sub-issues created
- Ticket status → Active (but no repair work to do)
- Skips straight to Admin's final confirmation
- When Admin confirms: ticket closes, vehicle stays Available, Issue Report marks Resolved (if linked)
- Total time: inspection + admin approval, no repair work

"Needs Maintenance" inspection:

- Custodian finds one or more defects
- Logs each defect as a separate sub-issue with title + maintenance type
- Ticket status → Active (repair mode engaged)
- Vehicle status → Under Maintenance
- For each sub-issue: Admin assigns mechanic → mechanic repairs → custodian verifies → Admin confirms
- Ticket only closes when all sub-issues are Done or Deferred
- Total time: days or weeks, multi-step workflow

Real scenario:

- **Scenario A:** Roel opens a ticket for Ambulance Doss because someone reported "brake issue". Nicole inspects and finds: "Actually, I tested the brakes and they're fine." She submits "No Issues". Ticket closes same day.
- **Scenario B:** Same ticket, but Nicole finds: "Brakes feel soft, pads worn." She submits "Needs Maintenance" with one sub-issue "Brake Pads". Now it's a days-long repair workflow.

Q15: Can an Admin directly inspect a ticket?

Answer: No. Only the assigned Custodian can inspect.

Why this matters: Inspection is a formal Tier-1 gate. Admin's job is to manage the overall process and make approvals, not to do field inspections. If Admin could bypass inspection, they could open a ticket and immediately assign mechanics without anyone physically verifying the vehicle. This breaks accountability.

If Admin needs to inspect: They must create a ticket, assign themselves (if they have Custodian role) or a Custodian, and have that person do the physical inspection. Two roles, two steps.

Q16: Why is deferring a sub-issue better than just deleting it?

Answer: Because deletion loses the record permanently; deferral creates a permanent breadcrumb.

If Admin deleted the sub-issue:

- Work is gone from the system
- Later, someone asks: "Why isn't the rear pads repair done?"
- Answer: "We don't know — it's not in the system anymore"
- Defect could be forgotten or overlooked

If Admin defers the sub-issue:

- Sub-issue → Deferred (immutable, visible in archives)
- Issue Report breadcrumb → Created, status Pending
- Anyone asking "Why aren't rear pads done?" can find the pending Issue Report
- Reason is documented: "Backordered, ETA July 30"
- Audit trail is complete

Deferral is an intentional decision to pause, not a deletion. It's recorded, auditable, and creates a guaranteed reminder.

SCENARIO-BASED QUESTIONS

SCENARIO 1

Walk through: A flat tire is discovered at readiness check time (right before dispatch).

Situation: It's 7:30 AM. Firetruck 2 is scheduled to go out on patrol at 8 AM. Mark runs the morning readiness check: all items Pass except "Tires" → one tire is flat.

System state:

- Readiness Check result: NOT READY (one item failed)
- Vehicle status: still Available (readiness ≠ vehicle status)
- Vehicle condition: Good (no damage reported yet)

What happens next:

1. **Mark calls Toto (mechanic):** "Firetruck 2 has a flat tire." Toto comes to the station with a new tire.
2. **Roel (Admin) has two choices:**
 - **Option A (Quick fix):** Toto changes the tire as a quick field repair. Roel logs a Maintenance Record directly (not a formal ticket) for "Tire Replacement — puncture repair". The record goes through Custodian verification (Mark test-drives) and Admin confirmation (Roel approves) and the vehicle returns to Available.
 - **Option B (Formal ticket):** Roel opens a Maintenance Ticket, assigns Mark to inspect, Mark inspects and logs the flat as a sub-issue, Roel assigns Toto, Toto repairs, Mark verifies, Roel confirms. Takes longer, full audit trail.
3. **Option A outcome (faster):** 15 minutes: tire changed, Maintenance Record logged, Mark verifies (test-drive), Roel confirms. Vehicle is back to Available/Good at ~7:50 AM. Patrol can start on time.
4. **Option B outcome (formal):** 1-2 hours: full ticket workflow. Patrol delayed.

Key insight: Readiness checks catch problems that would otherwise be discovered mid-response (when the vehicle is already in the field). The flat tire is now known at the station, where it can be fixed immediately, not 5 miles away on a call.

SCENARIO 2

Walk through: A mechanic discovers a secondary problem while working on a ticket (scope creep).

Situation: Ticket #1234 on Ambulance Doss has one sub-issue: "Brake Pads — wear". Toto (mechanic) is halfway through removing the old pads when he notices the master cylinder is weeping brake fluid — a secondary defect not in the original inspection.

Three options:

Option 1: Log both repairs in one work log

- Toto completes the brake pads + diagnoses the master cylinder
- Logs one repair: "Replaced brake pads + replaced master cylinder"
- Cost covers both, one Maintenance Record
- **Problem:** Loses granularity. If the master cylinder repair fails later, which one is to blame?

Option 2: Add a sub-issue mid-ticket

- Toto continues with brake pads, logs completion
- Then adds a new sub-issue to the same ticket: "Master Cylinder — fluid leak" (if system allows, only Custodian or assigned mechanic can add)
- Admin assigns a mechanic to the new sub-issue (could be Toto again or someone else)
- Master cylinder work is tracked independently
- Two separate Maintenance Records, clear accountability
- **Advantage:** Ticket captures both defects, full audit trail

Option 3: Complete current sub-issue, create new ticket later

- Toto completes brake pads, Nicole verifies, Roel confirms
- Ticket #1234 closes
- Roel creates a new ticket #1235 specifically for the master cylinder
- Repeats the workflow: inspect, assign mechanic, repair, verify, confirm
- **Advantage:** Two tickets, no scope creep. Cleaner separation.

Best practice: Option 2 (add mid-ticket) keeps related work on the same ticket, preserving context. If the master cylinder is unrelated, Option 3 (new ticket) is cleaner.

SCENARIO 3

Walk through: A mechanic becomes unavailable mid-repair (gets sick, transferred).

Situation: Ticket #1234 has two sub-issues: Front Brakes (Done), Rear Brakes (Under Repair, assigned to Toto). Toto has already logged partial work: "Removed old pads, waiting for new pads to arrive." On day 3, Toto calls in sick and won't be back for a week.

What Roel (Admin) does:

1. Opens the Rear Brakes sub-issue → Reassign Mechanic
2. Selects Marco (different mechanic)
3. Provides reason: "Original mechanic (Toto) unavailable due to illness. Reassigned to Marco."
4. System updates: sub-issue assigned to Marco, both mechanics notified
5. Sub-issue status: stays Under Repair (no change)

What Marco sees:

- Work Order assigned to him: "Rear Brakes — Ambulance Doss"
- He can see Toto's partial work log: "Removed old pads, awaiting new pads"
- New pads are in stock at the station
- Marco installs them, bleeds the system, completes the work
- Logs his completion: "Installed new rear pads, bled system, ready for verification"

Final audit trail shows: Toto did partial work, Marco completed it. Full context preserved.

Alternative (if no replacement mechanic): Roel could defer the sub-issue temporarily: "Mechanic reassignment pending." The ticket can close with deferred work, vehicle returns to Available (marked "Not Fit"), and when another mechanic becomes available, Roel reassigns or creates a new ticket.

SCENARIO 4

Walk through: A Custodian's inspection finds "No Issues", but Admin wants to double-check.

Situation: Ticket #1234 for Ambulance Doss. Nicole inspects and finds no defects → submits "No Issues". Ticket status → Active. Roel (Admin) reviews the report and is skeptical: "The original Issue Report said brakes felt weak. Did she really test-drive it?"

What can Roel do?

Option 1: Cancel the ticket and reassign to a different custodian

- Roel cancels the ticket
- Creates a new ticket for the same vehicle/defect
- Assigns a different custodian (Mark) for a second opinion
- Mark inspects and submits his findings

Option 2: Don't approve yet, request re-inspection

- Roel sees the "No Issues" submission but hasn't clicked Confirm yet
- He can reopen/reject the inspection and ask Nicole: "Can you test-drive again and confirm brakes?"
- If system allows, Roel could annotate: "Requires verification" and return to Nicole
- Nicole retests, confirms again

Option 3: Confirm but schedule a follow-up readiness check

- Roel trusts Nicole's inspection, confirms the ticket
- But schedules a readiness check for Ambulance Doss within a week
- At that time, Mark can include a brake inspection in the readiness check
- If brakes are still soft, a new ticket can be opened

Key point: Admin has limited visibility into whether the Custodian actually tested the vehicle. That's why the attestation checkbox is important — it's a signed affidavit. If Roel doubts the inspection, he can cancel the ticket and request a second opinion, or schedule a readiness check to double-check.

SCENARIO 5

Walk through: A ticket must close but work is unfinished (Decision Close with "Not Fit").

Situation: Ticket #1234 on Firetruck 2 has three sub-issues: Hoses (Done), Pump (Done), Tank Seal (Under Repair, stalled waiting for a replacement seal from the manufacturer, ETA 2 weeks). It's July 23. Budget/funds for July are exhausted. Roel decides to close the ticket and reopen in August.

Roel opens the ticket → Close Ticket button is disabled (because Tank Seal is still Under Repair, not Done).

He clicks "Decision Close" instead:

1. System prompts: "Cannot close with unfinished sub-issues. Reason for closure?"
2. Roel enters: "Budget exhausted for July. Will reopen as new ticket in August when funds available."
3. System asks: "Is Firetruck 2 fit for service?"
4. Roel answers: "No — Tank Seal is still leaking, vehicle cannot hold water for fire response."
5. Confirm

System automatically:

- Ticket → Closed (moved to Ticket Archives)
- Tank Seal sub-issue → Deferred (with Roel's reason in record)
- Issue Report breadcrumb → Created: "Tank Seal — deferred from Ticket #1234" status = Pending
- Vehicle status: Under Maintenance (stays locked, cannot dispatch)

What happens in August:

- Seal arrives
- Roel opens Issue Reports, finds the Tank Seal breadcrumb
- Clicks "Create Ticket" from the breadcrumb
- New Ticket #1235 is created specifically for tank seal repair
- Normal workflow resumes: inspect, assign mechanic, repair, verify, confirm
- Firetruck 2 finally returns to Available

Why "Not Fit"?: Because Roel explicitly decided the vehicle cannot be dispatched until the seal is fixed. If he had answered "Fit for service", the vehicle would return to Available, but the breadcrumb Issue Report would still exist as a record of the deferred work.

SCENARIO 6

Walk through: Multiple vehicles of the same type are Under Maintenance simultaneously (fragility alert).

Situation: Barangay has 2 ambulances: Ambulance Doss and Ambulance Zero. Both are critical for emergency response.

Timeline:

Monday: Ambulance Doss starts a maintenance ticket (brake repair). Status → Under Maintenance. Ambulance Zero is Available/Good/Ready. Dashboard readiness for Ambulances: 1/2 (Ready). Badge = "Ready, but fragile — one ambulance away from zero coverage."

Wednesday: Someone files an Issue Report for Ambulance Zero (overheating). Roel creates a ticket and assigns inspection to Nicole. Ambulance Zero now Under Maintenance. Ambulances readiness: 0/2. Dashboard badge = "NO COVERAGE — Critical."

Red flag notification: System alerts Roel: "Ambulance type is at zero coverage. All units under maintenance or out of service."

What Roel does:

- Calls both mechanics: "Prioritize both ambulance repairs. We have zero ambulances available."
- Escalates budget requests if needed to get parts faster
- Considers deferring non-critical sub-issues to get one ambulance back in service faster
- Checks with Custodian: "Can we close Doss's ticket early if we mark it 'Not Fit' and defer the minor issue?"

Outcome: Doss completes brake repair (Done), Roel decides to close it and defer a minor hose inspection to next month. Doss returns to Available. Ambulances readiness: 1/2 (no longer critical).

Meanwhile: Zero's overheating is diagnosed as a radiator issue. New radiator on order. Will take a week. Roel defers that sub-issue too, marks ticket closed as "Not Fit" to preserve the defect record.

One week later: Radiator arrives, new ticket created from the breadcrumb, Zero is repaired, back to Available. Ambulances readiness: 2/2 (fully staffed).

Key insight: The fragility alert on the dashboard made Roel aware of the critical situation and drove fast decision-making. Without the alert, both ambulances might have stayed Under Maintenance for weeks with no sense of urgency.

SCENARIO 7

Walk through: A Custodian rejects a repair because the vehicle doesn't meet specifications.

Situation: Ticket #1234 on Firetruck 2. Sub-issue: "Pump Replacement". Toto replaced the pump with a new OEM pump. Logs repair: "Replaced pump per specification." Nicole (Custodian) receives verification notification.

Nicole test-drives and runs the functional checklist:

- Pump primes: FAIL — the new pump doesn't prime correctly. System is not building full pressure.
- Water flow insufficient: FAIL — not meeting the flow specification

Nicole submits verdict: REJECTED with notes: "New pump doesn't meet flow specs. Priming is weak. Needs investigation — possible bad unit or wrong model installed."

System response:

- Sub-issue → Back to Under Repair
- Toto's repair log is deleted/superseded
- Toto is notified: "Repair rejected by Custodian Nicole. Notes: New pump doesn't meet flow specs."

What Toto does:

1. Investigates the pump: tests it in isolation, confirms it's not pumping to spec
2. Checks the model number against the specification sheet — he realizes it's the wrong model
3. Orders the correct model
4. Removes the incorrect pump, installs the correct one
5. Tests in the vehicle, confirms water flow is now correct
6. Logs new repair: "Removed incorrect pump model [wrong model], installed correct model [correct model] per specification. System now primes correctly and meets flow spec."

Nicole verifies again:

- Pump primes: PASS

- Water flow: PASS (meets spec)
- All other items: PASS
- Approves

Roel confirms: Ticket sub-issue → Done.

Audit trail: Shows first pump installation (rejected), new pump installation (approved). Full visibility into what went wrong and how it was fixed.

SCENARIO 8

Walk through: An Issue Report is created, but Admin decides not to open a ticket (false alarm).

Situation: Nicole files an Issue Report: "Ambulance Doss battery feels weak — slow cranking on Wednesday morning." Roel (Admin) reviews it and calls Nicole to get more details.

Phone conversation: Nicole explains: "It was 6 AM, car was cold, I didn't think much of it. But I haven't checked the battery voltage." Roel decides to verify before opening a ticket: "Let me have Mark do a quick check."

Mark arrives, tests the battery: 13.5V (full charge). Starter spins fine. No issue detected. It was likely just cold-start behavior.

Roel's decision: "No formal ticket needed. It was a false alarm." He closes/deletes the Issue Report.

What happens:

- Issue Report → Deleted (or marked Closed without creating a ticket)
- Vehicle status: unchanged
- No Maintenance Ticket ever created
- No Maintenance Record
- Audit trail shows: Issue Report #[ID] created by Nicole, reviewed by Roel, closed as false alarm

Contrast this to a real issue: If the battery was actually weak (10V), Roel would have created a ticket, Toto would replace the battery, Nicole would verify it starts fine, Roel would confirm, and a Maintenance Record would be created as proof.

Key insight: Not every Issue Report becomes a ticket. Admin acts as a gate to filter out false alarms and trivial issues, saving the team from unnecessary formal processes.

SCENARIO 9

Walk through: Preventive maintenance (scheduled oil change) is completed on schedule and auto-recurs.

Situation: Roel creates a maintenance schedule: Firetruck 2, Oil Change, July 23, 2026, recurring every 6 months.

July 23, 2026 (scheduled day):

1. Toto (mechanic) performs the oil change: drains old oil, installs new OEM oil, changes filter
2. Opens Maintenance Schedule → Find "Firetruck 2 Oil Change" scheduled for today → "Mark Done"
3. Fills: cost (\$65), notes ("Changed Mobil 5W-30, 20L, installed OEM filter")
4. Confirms

System automatically:

- Schedule → Completed
- Maintenance Record → Created with Toto's data (cost, notes, date)
- Next scheduled event → Auto-created for January 23, 2027 (6 months later)

Dashboard: The Preventive Maintenance Watch list no longer shows the July 23 oil change (it's Completed). But shows the January 23 oil change as "Due in 6 months."

January 23, 2027 (six months later):

- Toto (or another mechanic) performs the oil change again
- Marks the January schedule Done
- Maintenance Record created
- Next schedule auto-created for July 23, 2027

No breaks in the chain: The system ensures preventive maintenance is never forgotten. Each completion triggers the next recurrence automatically.

If the chain breaks: If someone deletes the January schedule without completing it, the system won't auto-create the next one. Admin would have to manually create a new schedule. That's why the recurring logic only triggers on Mark Done, not on deletion.

SCENARIO 10

Walk through: A vehicle is decommissioned due to age/damage, later needed again.

Situation: Ambulance Zero has been unreliable for months — constant electrical issues, repair costs mounting. Roel decides it's time: end-of-life writeoff.

Decommissioning process:

1. First, Roel closes any open tickets on Ambulance Zero (cannot decommission with active tickets)
2. Opens Vehicle Management → Ambulance Zero → Decommission
3. Provides reason: "Beyond economical repair. Cumulative maintenance costs exceed 70% of replacement value. Recommend decommissioning."
4. Confirms

System response:

- Vehicle status → Decommissioned
- Removed from active vehicle pickers (not available for new tickets)
- Removed from dashboard readiness calculations
- Removed from vehicle type coverage panels
- History preserved in archives

Three months later (October 2026): Barangay gets budget approval to repair Ambulance Zero (new engine and wiring harness approved for ~\$8,000). Roel decides to bring it back.

Restoration:

1. Vehicle Management → Ambulance Zero → Restore
2. Status → Available / Good
3. Reappears in vehicle pickers, dashboard, readiness calculations

Then Roel creates a ticket: "Engine Replacement + Electrical Overhaul" to address the issues that caused the decommissioning.

After that ticket is done: Ambulance Zero is fully operational again and back in active service.

Audit trail: Shows decommissioning reason, restoration date, and full repair history. No loss of information.

Barangay VMS Defense Q&A | Tricky Questions, Edge Cases & Scenarios

Comprehensive reference for capstone project defense discussions. All scenarios tested in development and QA.